COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S3
DATE PERFORMED:	10/29/2026	**DATE SUBMITTED:**	10/29/2026
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Exam	**Final Hands-on Skills Exam**

1. PROCEDURE OUTPUT

Github Repository: https://github.com/piosjoshua/Final_Exam_Pios.git

Chosen Enterprise Service: **NGINX**

- I researched NGINX and found out that it is used to host websites and serve web pages to users. I chose it for my project because I wanted to deploy a website on both Ubuntu and CentOS.

Chosen Monitoring Tool: **Uptime Kuma**

- I researched Uptime Kuma and found out that it is used to monitor websites and check if they are online or offline. I chose it for my project so I could easily check if my websites were working properly.

Figure 1. Project Directory and File Structure

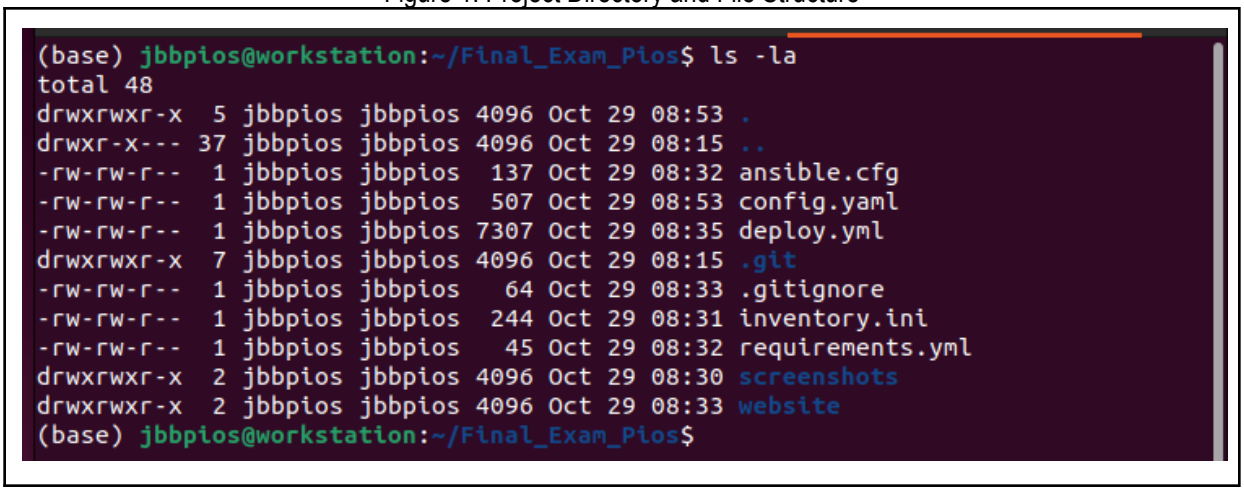


Figure 2. Ansible Controller Configuration

```
(base) jbbpios@workstation:~/Final_Exam_Pios$ cat ansible.cfg

[defaults]
inventory = inventory.ini
host_key_checking = True
retry_files_enabled = False
interpreter_python = auto_silent
timeout = 30
```

Figure 3. Inventory of Ubuntu and CentOS

```
(base) jbbpios@workstation:~/Final_Exam_Pios$ cat inventory.ini

[ubuntu]
ubuntu-vm1 ansible_host=192.168.56.101 ansible_user=jppios

[centos]
centos-vm ansible_host=192.168.56.104 ansible_user=jpios

[enterprise:children]
ubuntu
centos

[monitoring:children]
ubuntu
centos

[managed:children]
ubuntu
centos
```

Figure 4. Configuration of config.yaml

```
(base) jbbpios@workstation:~/Final_Exam_Pios$ cat config.yaml

---
project_name: Final_Exam_Pios

managed_username: jbbpios
motd_message: "Ansible Managed by jbbpios"

enterprise:
  name: nginx-enterprise
  image: nginx:alpine
  host_port: 8081
  container_port: 80

monitoring:
  name: uptime-kuma
  image: louislam/uptime-kuma:2
  host_port: 3001
  container_port: 3001
  data_volume: uptime-kuma-data

monitor_targets:
```

Figure 5. Playbook for automated service deployment

```
(base) jbbpios@workstation:~/Final_Exam_Pios$ cat deploy.yml

---
- name: Prepare Ubuntu and CentOS servers
  hosts: managed
  become: true
  gather_facts: true
  vars_files:
    - config.yaml

  tasks:
    - name: Validate operating system
      ansible.builtin.assert:
        that:
          - ansible_distribution in ['Ubuntu', 'Debian', 'CentOS']
        fail_msg: "Only Ubuntu, Debian and CentOS are supported."

    - name: Check whether Docker is installed
      ansible.builtin.command: docker --version
      register: docker_check
      changed_when: false
      failed_when: false
```

Figure 6. Validation of playbook syntax

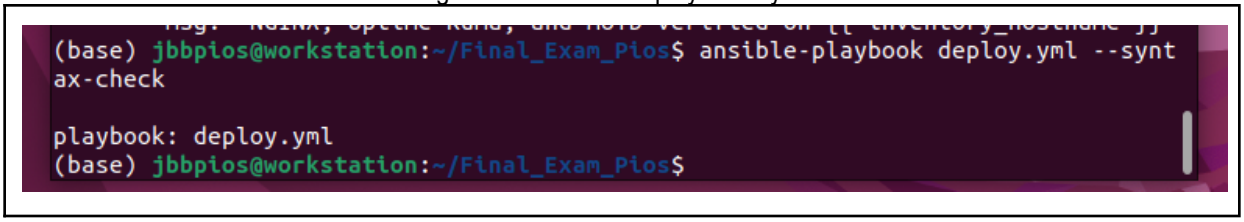


Figure 7. Ansible connectivity test

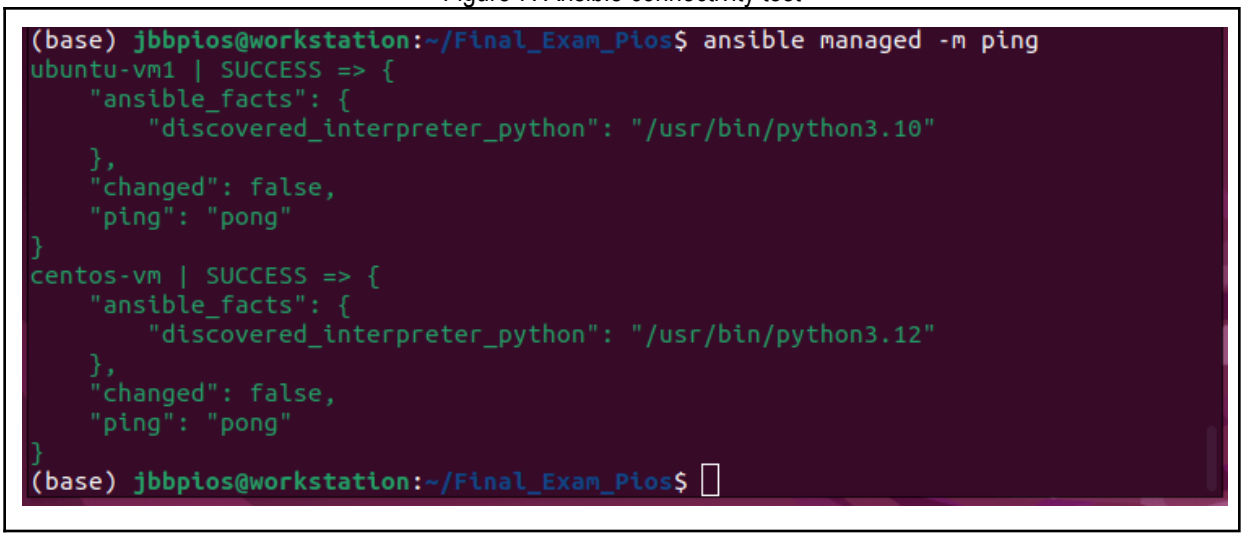


Figure 8. Execution of playbook

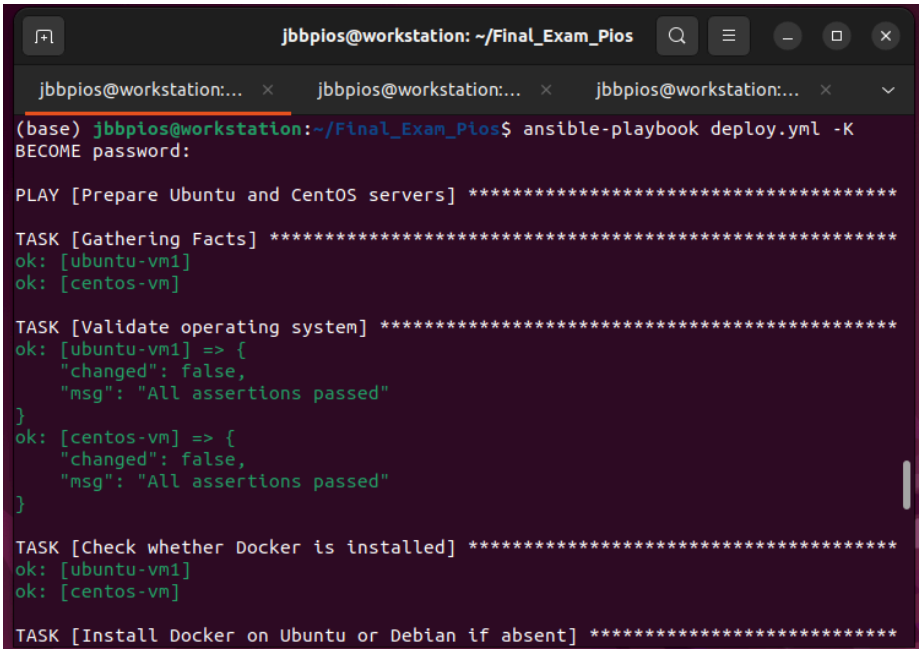


Figure 9. Successful Deployment summary

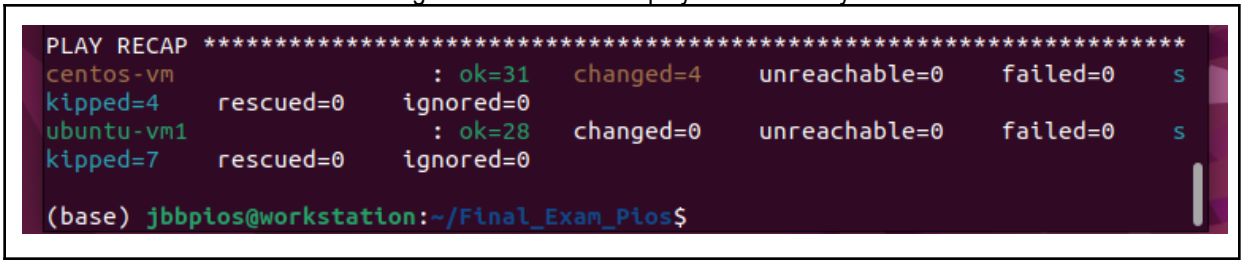


Figure 10. Containers of Ubuntu

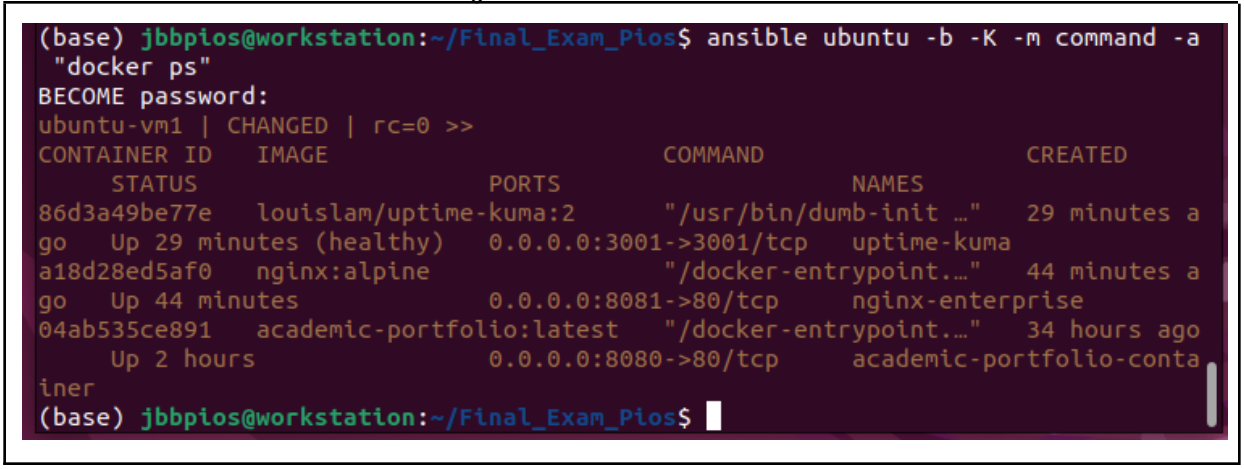


Figure 11. Containers on CentOS

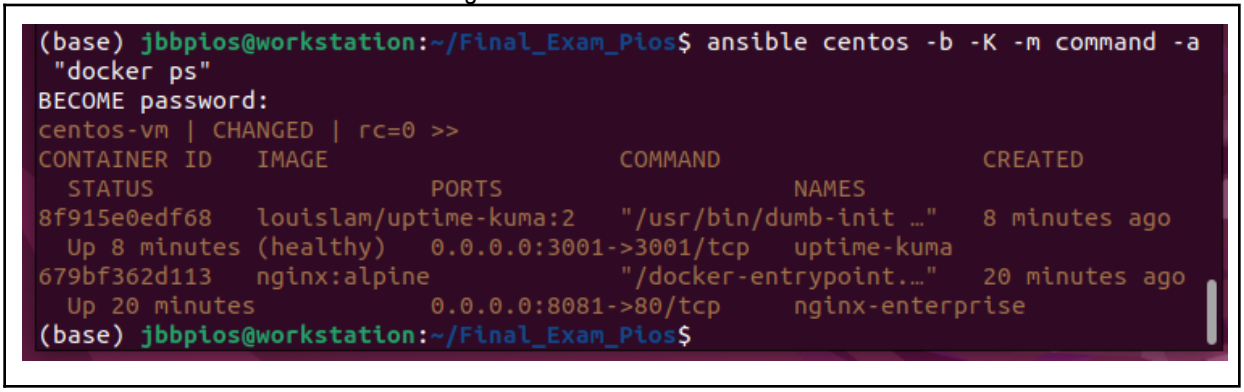


Figure 12. Enterprise Service on Ubuntu

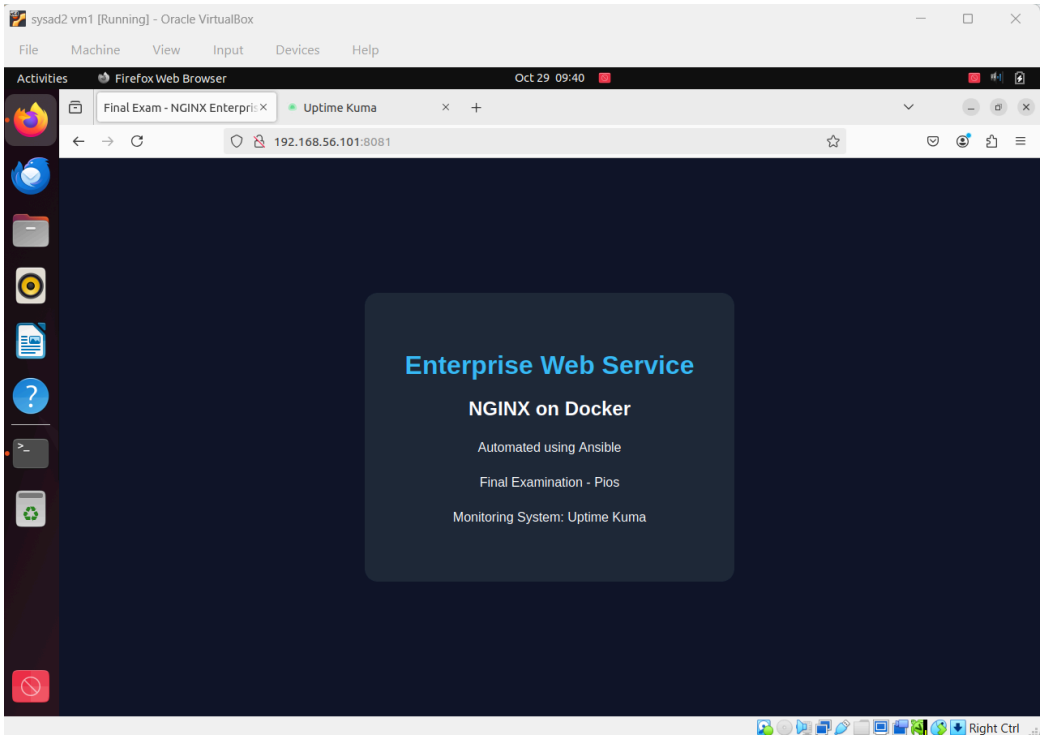


Figure 13. Enterprise Service on CentOS

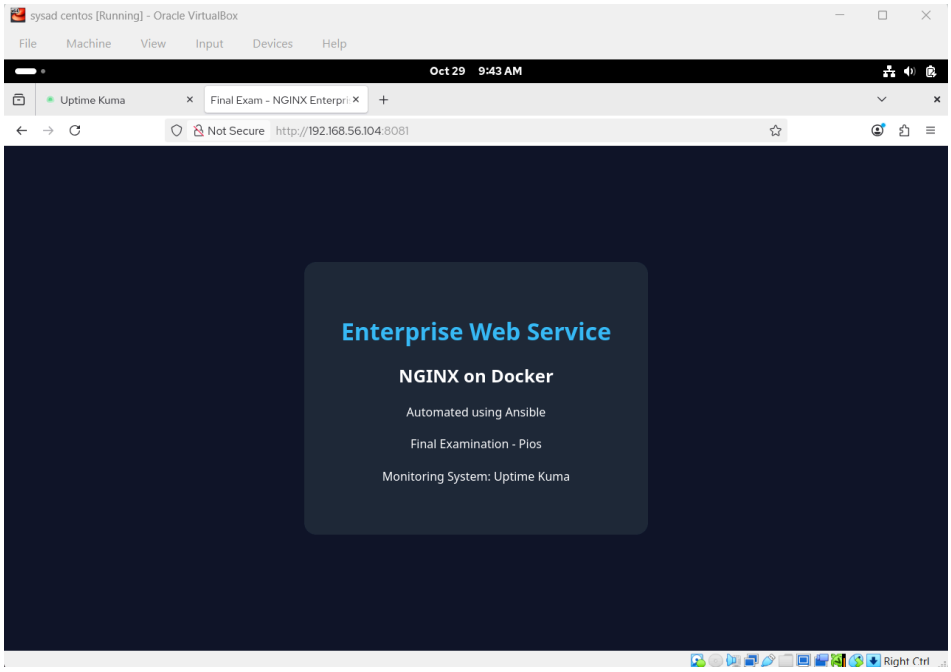


Figure 14. Http service verification

```
(base) jbbpios@workstation:~/Final_Exam_Pios$ curl -I http://192.168.56.104:8081
HTTP/1.1 200 OK
Server: nginx/1.31.8
Date: Thu, 29 Oct 2026 01:43:18 GMT
Content-Type: text/html
Content-Length: 830
Last-Modified: Thu, 29 Oct 2026 00:42:57 GMT
Connection: keep-alive
ETag: "6ae29691-33e"
Accept-Ranges: bytes

(base) jbbpios@workstation:~/Final_Exam_Pios$
```

Figure 15. Monitoring Tool on Ubuntu

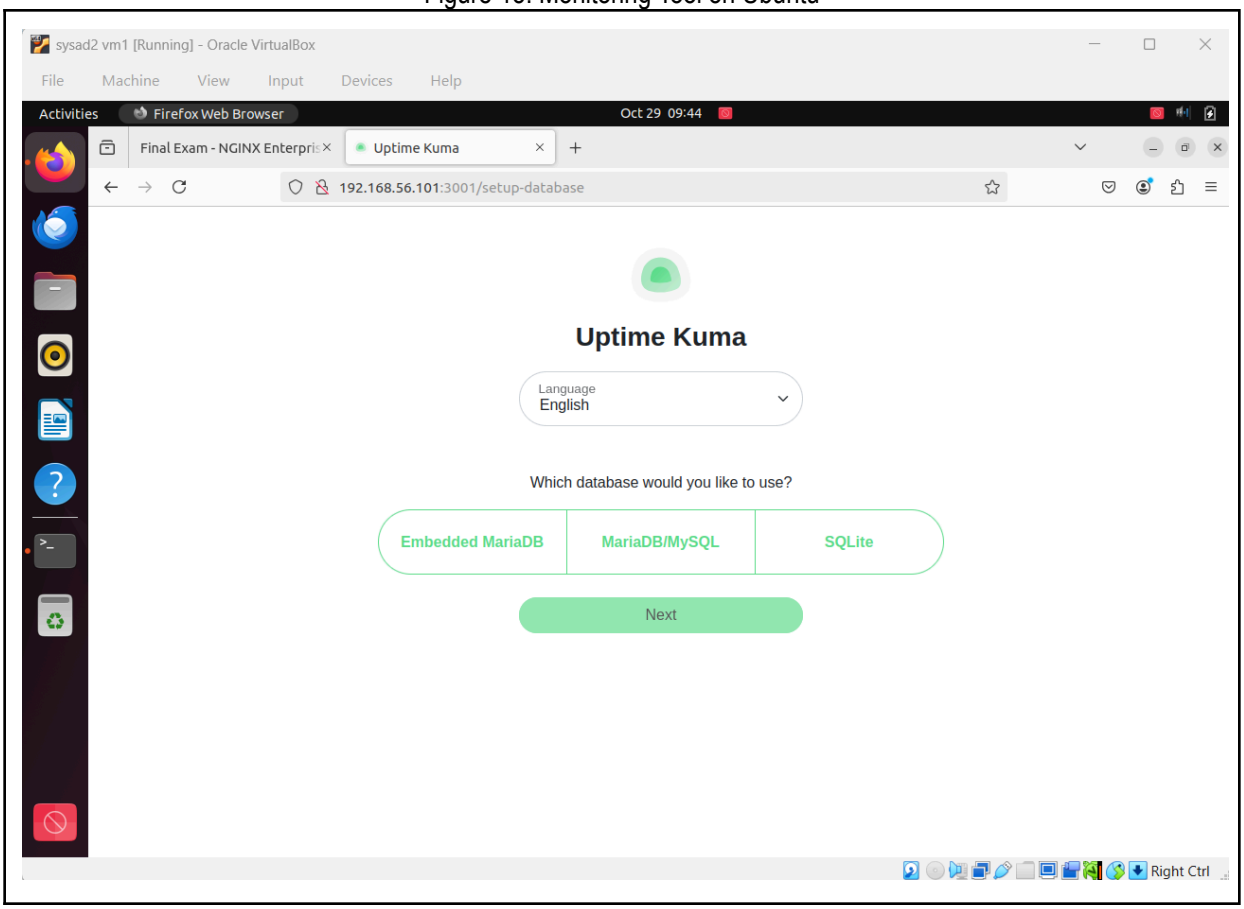


Figure 16. Monitoring on CentOS

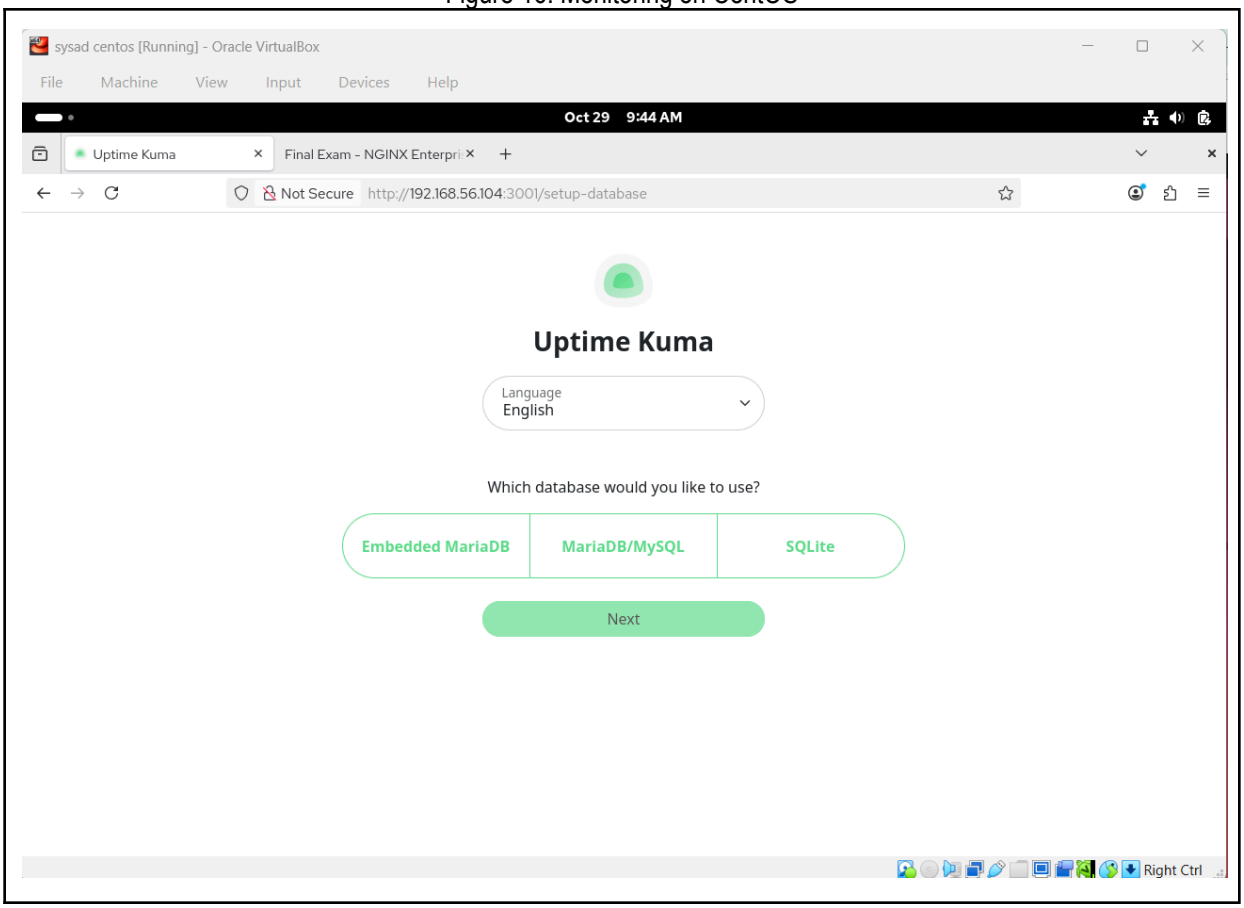
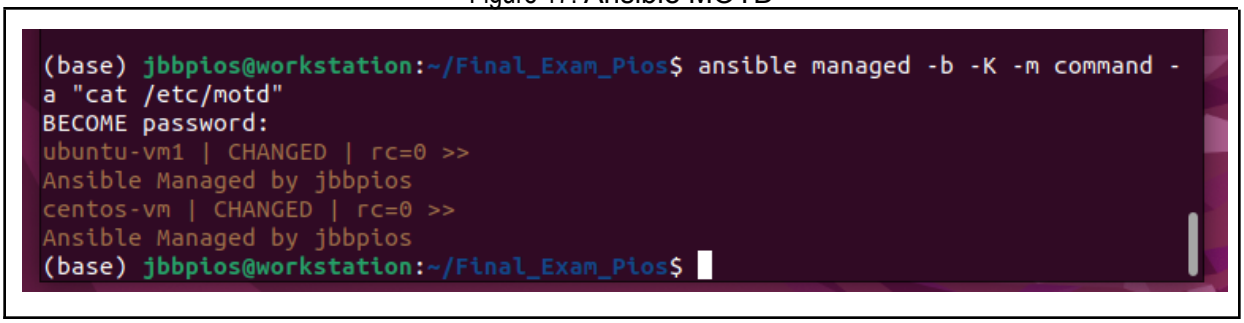


Figure 17. Ansible MOTD



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ansible.cfg

```
[defaults]
inventory = inventory.ini
host_key_checking = True
retry_files_enabled = False
interpreter_python = auto_silent
timeout = 30
```

config.yaml

```
---
project_name: Final_Exam_Pios

managed_username: jbbpios
motd_message: "Ansible Managed by jbbpios"

enterprise:
  name: nginx-enterprise
  image: nginx:alpine
  host_port: 8081
  container_port: 80

monitoring:
  name: uptime-kuma
  image: louislam/uptime-kuma:2
  host_port: 3001
  container_port: 3001
  data_volume: uptime-kuma-data

monitor_targets:
  - name: Ubuntu Enterprise Server
    url: "http://192.168.56.101:8081"

  - name: CentOS Enterprise Server
    url: "http://192.168.56.104:8080"
```

deploy.yml

```
---
- name: Prepare Ubuntu and CentOS servers
  hosts: managed
  become: true
  gather_facts: true
  vars_files:
    - config.yaml
```

```
tasks:
  - name: Validate operating system
    ansible.builtin.assert:
      that:
        - ansible_distribution in ['Ubuntu', 'Debian', 'CentOS']
      fail_msg: "Only Ubuntu, Debian and CentOS are supported."

  - name: Check whether Docker is installed
    ansible.builtin.command: docker --version
    register: docker_check
    changed_when: false
    failed_when: false

  - name: Install Docker on Ubuntu or Debian if absent
    ansible.builtin.apt:
      name:
        - docker.io
        - curl
        - python3
      state: present
      update_cache: true
    when:
      - ansible_os_family == 'Debian'
      - docker_check.rc != 0

  - name: Install CentOS repository utility
    ansible.builtin.dnf:
      name: dnf-plugins-core
      state: present
    when:
      - ansible_distribution == 'CentOS'
      - docker_check.rc != 0

  - name: Add official Docker repository on CentOS
    ansible.builtin.get_url:
      url: https://download.docker.com/linux/centos/docker-ce.repo
      dest: /etc/yum.repos.d/docker-ce.repo
      mode: "0644"
    when:
      - ansible_distribution == 'CentOS'
      - docker_check.rc != 0

  - name: Install Docker Engine on CentOS
    ansible.builtin.dnf:
      name:
        - docker-ce
        - docker-ce-cli
        - containerd.io
```

```
    state: present
  when:
    - ansible_distribution == 'CentOS'
    - docker_check.rc != 0

- name: Enable and start Docker
  ansible.builtin.systemd:
    name: docker
    state: started
    enabled: true

- name: Verify Docker daemon
  ansible.builtin.command: docker info
  changed_when: false

- name: Set required MOTD
  ansible.builtin.copy:
    content: "{{ motd_message }}\n"
    dest: /etc/motd
    owner: root
    group: root
    mode: "0644"

- name: Check whether firewalld is active
  ansible.builtin.command: systemctl is-active firewalld
  register: firewalld_check
  failed_when: false
  changed_when: false
  when: ansible_distribution == 'CentOS'

- name: Allow HTTP application ports on CentOS
  ansible.builtin.command:
    argv:
      - firewall-cmd
      - --permanent
      - "--add-port={{ item }}/tcp"
  loop:
    - "{{ enterprise.host_port }}"
    - "{{ monitoring.host_port }}"
  register: firewall_result
  changed_when: "'ALREADY_ENABLED' not in (firewall_result.stdout | upper)"
  when:
    - ansible_distribution == 'CentOS'
    - firewalld_check.stdout == 'active'

- name: Reload CentOS firewall
  ansible.builtin.command: firewall-cmd --reload
  changed_when: true
```

```
      when:
        - ansible_distribution == 'CentOS'
        - firewalld_check.stdout == 'active'

- name: Deploy NGINX Enterprise Web Service
  hosts: enterprise
  become: true
  gather_facts: true
  vars_files:
    - config.yaml

  tasks:
    - name: Create NGINX website directory
      ansible.builtin.file:
        path: /opt/final-exam-nginx
        state: directory
        mode: "0755"

    - name: Copy enterprise webpage
      ansible.builtin.copy:
        src: website/index.html
        dest: /opt/final-exam-nginx/index.html
        mode: "0644"
      register: website_copy

    - name: Deploy NGINX Docker container
      community.docker.docker_container:
        name: "{{ enterprise.name }}"
        image: "{{ enterprise.image }}"
        state: started
        pull: missing
        restart_policy: unless-stopped
        recreate: "{{ website_copy.changed }}"
        published_ports:
          - "{{ enterprise.host_port }}:{{ enterprise.container_port }}"
        volumes:
          - "/opt/final-exam-nginx/index.html:/usr/share/nginx/html/index.html:ro{{ ',Z' if ansible_distribution ==
'CentOS' else '' }}"

    - name: Verify NGINX HTTP response
      ansible.builtin.uri:
        url: "http://127.0.0.1:{{ enterprise.host_port }}/"
        status_code: 200
        return_content: true
      register: nginx_result
      retries: 10
      delay: 3
      until: nginx_result.status == 200
```

```
    - name: Check enterprise webpage content
      ansible.builtin.assert:
        that:
          - "'Enterprise Web Service' in nginx_result.content"
        success_msg: "NGINX verified on {{ inventory_hostname }}."

- name: Deploy Uptime Kuma Monitoring
  hosts: monitoring
  become: true
  vars_files:
    - config.yaml

  tasks:
    - name: Create persistent monitoring volume
      community.docker.docker_volume:
        name: "{{ monitoring.data_volume }}"
        state: present

    - name: Start Uptime Kuma container
      community.docker.docker_container:
        name: "{{ monitoring.name }}"
        image: "{{ monitoring.image }}"
        state: started
        pull: missing
        restart_policy: unless-stopped
        published_ports:
          - "{{ monitoring.host_port }}:{{ monitoring.container_port }}"
        volumes:
          - "{{ monitoring.data_volume }}:/app/data"

    - name: Verify Uptime Kuma dashboard
      ansible.builtin.uri:
        url: "http://127.0.0.1:{{ monitoring.host_port }}/"
        status_code: 200
      register: monitoring_result
      retries: 24
      delay: 5
      until: monitoring_result.status == 200

    - name: Display monitoring configuration
      ansible.builtin.debug:
        msg:
          - "Monitoring installed on {{ inventory_hostname }}"
          - "Configure monitors for {{ monitor_targets | map(attribute='url') | list }}"

- name: Verify both services and MOTD
  hosts: managed
```

```
become: true
vars_files:
  - config.yaml

tasks:
  - name: Inspect NGINX
    community.docker.docker_container_info:
      name: "{{ enterprise.name }}"
    register: nginx_info

  - name: Inspect Uptime Kuma
    community.docker.docker_container_info:
      name: "{{ monitoring.name }}"
    register: monitoring_info

  - name: Confirm both containers are running
    ansible.builtin.assert:
      that:
        - nginx_info.exists
        - nginx_info.container.State.Running
        - monitoring_info.exists
        - monitoring_info.container.State.Running

  - name: Verify NGINX endpoint
    ansible.builtin.uri:
      url: "http://127.0.0.1:{{ enterprise.host_port }}/"
      status_code: 200

  - name: Verify monitoring endpoint
    ansible.builtin.uri:
      url: "http://127.0.0.1:{{ monitoring.host_port }}/"
      status_code: 200

  - name: Read configured MOTD
    ansible.builtin.slurp:
      src: /etc/motd
    register: motd_data

  - name: Verify MOTD is correct
    ansible.builtin.assert:
      that:
        - (motd_data.content | b64decode | trim) == motd_message

  - name: List running containers
    ansible.builtin.command: docker ps
    changed_when: false
    register: containers
```

```
  - name: Display container information
    ansible.builtin.debug:
      var: containers.stdout_lines

  - name: Display successful verification
    ansible.builtin.debug:
      msg: "NGINX, Uptime Kuma, and MOTD verified on {{ inventory_hostname }}"
```

inventory.ini

```
[ubuntu]
ubuntu-vm1 ansible_host=192.168.56.101 ansible_user=jppios

[centos]
centos-vm ansible_host=192.168.56.104 ansible_user=jpios

[enterprise:children]
ubuntu
centos

[monitoring:children]
ubuntu
centos

[managed:children]
ubuntu
centos
```

requirements.yml

```
---
collections:
  - name: community.docker
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: N/A

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: N/A

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: N/A

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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